DO NOW — RECALL

Lesson 052 · Binary Search Trees with Node Objects

Name: _____ Class: _____ Date: _____

1. What does LinkedList<T>.AddAfter use to place a new node?

2. What does HEAD mean for a linked list?

3. What is an invariant of a BST?

4. A binary search tree has root 10, left child 6 and right child 18. Insert 15, then 12, using the usual smaller-left, larger-right rule. Determine the parent and side of the new node 12.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

An existing node reference.

2. Model answer

Its first node.

3. Model answer

Left subtree keys less and right keys greater under chosen policy.

4. Model answer

12 is the left child of 15. Its search path is right from 10, left from 18, then left from 15.

By the end, you can…

• A binary search tree (BST) keeps smaller keys in the left subtree and larger keys in the right subtree under a consistent duplicate-key policy.

• Each node stores a key and optional Left/Right child references. Inserting begins at the root and follows comparisons until an empty child link is found.

• Inorder traversal (left, root, right) prints sorted keys for a valid BST without duplicates.

Vocabulary — know these terms

BST — Binary search tree ordered by key.

root — Topmost tree node.

leaf — Node without child nodes.

inorder — Left subtree, root, right subtree.

NEW LEARNING

Read, annotate and explain

Discover

A binary search tree (BST) keeps smaller keys in the left subtree and larger keys in the right subtree under a consistent duplicate-key policy.

Trace

Each node stores a key and optional Left/Right child references. Inserting begins at the root and follows comparisons until an empty child link is found.

Evaluate

Inorder traversal (left, root, right) prints sorted keys for a valid BST without duplicates.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Insert key 7 into BST root 8, left 3, right 10, with 6 right of 3.

Explain the trace

7 becomes right child of 6; follow comparisons until a null child is reached.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

BST root8, left3, right10, insert 6. Show comparisons and location. [6]

Question 2

For BST with keys 8,3,10,6, give inorder traversal and explain sorting. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why BST search may be O(log n) when balanced but O(n) when skewed. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An unbalanced BST is guaranteed to have logarithmic search time.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.